DGMS 2019 • SECTION G: SEISMOLOGY & EARTH INTERIOR

Question 61: Paleoseismicity & Field Evidence of Past Fault Slippage

Complete Diagnostic Analysis, Structural Field Methods, and Reference Textbook Citations

OFFICIAL EXAMINATION QUESTION

Palaeoseismicity is related to the study of

(a) Earthquakes of the geological past through slippage observed in the field along faults [CORRECT]

(b) Relationship between earthquake and faulting

(c) Mapping of earthquakes

(d) Risk related to seismic activities

VERIFIED ANSWER KEY

Correct Option: (a) — *Palaeoseismicity (Paleoseismology)* is defined scientifically as the geological investigation of ancient, prehistoric earthquakes through the direct observation and dating of coseismic fault displacement, surface rupture, colluvial wedges, distorted strata, and liquefaction structures preserved along active faults in the field.

1. Geological Definition & Scope of Paleoseismology

Paleoseismology (or *palaeoseismicity*) is an interdisciplinary branch of structural geology, geomorphology, and seismology dedicated to deciphering the location, timing, and size of prehistoric earthquakes that occurred prior to the advent of instrumental seismographs (installed globally primarily after World War II) and written historical chronicles.

While instrumental seismology covers roughly the past century and historical records cover a few hundred to thousands of years, the recurrence intervals of large, high-magnitude plate-boundary or intraplate earthquakes often span hundreds to tens of thousands of years.

Paleoseismology bridges this temporal gap by examining the physical rock and sediment record for fossilized signatures of ground rupture and ground shaking.



FIGURE 8.37 Nearly horizontal layers of light-colored rock and gravel, and dark-colored peat have been broken and distorted as a result of earthquake activity near Banning, California, along the San Andrea Fault.

suffered earthquakes in the past. Networks of seismo- stress release associated with sudden mineral phase

Figure 61.1 | Paleoseismic Trench Evidence of Fault Slip: Nearly horizontal layers of light-colored rock, gravel, and dark organic peat broken, distorted, and displaced across active fault strands near Banning, California, along the San Andreas Fault system. Such cross-cutting relationships and displaced Holocene stratigraphy allow structural geologists to reconstruct individual prehistoric rupture events and calculate earthquake recurrence intervals. (Source: Van der Pluijm & Marshak, *Earth Structure*, 2nd Ed., Fig. 8.37, p. 200).

2. Key Field Criteria & Paleoseismic Methodology

As documented by **Van der Pluijm & Marshak (2004, Chapter 8, pp. 200–201)**, paleoseismologists utilize explicit field techniques to identify ancient seismic slip:

- **Paleoseismic Trenching across Fault Scarps:** Excavating exploratory trenches perpendicular to active fault traces exposes stratigraphic horizons (colluvial wedges, paleosols, peat beds, tephra layers) offset by episodic fault movements. Successive upward terminations of fault splays date individual slip events.
- **Colluvial Wedges:** When surface faulting creates a scarp, weathering and gravity rapidly shed a wedge-shaped debris deposit (colluvial wedge) at the scarp base. Burying this wedge beneath subsequent flat sedimentation marks a distinct paleo-earthquake event.

- **Seismites & Paleoliquefaction:** Violent seismic shaking causes pore-water pressurization in saturated cohesionless sand layers, producing clastic dikes, sand blows (sand volcanoes), and convolute bedding. Radiocarbon dating ($\mathrm{^{14}C}$) of organic material in or adjacent to liquefied beds yields the earthquake date.
- **Offset Geomorphic Markers:** Displaced stream channels, offset alluvial fans, and offset glacial moraines reveal cumulative displacement, allowing slip-rate calculations ($\text{Slip Rate} = \text{Displacement} / \text{Age}$).
- **Pseudotachylytes:** Glassy, frictionally melted rocks formed along fault surfaces during high-velocity coseismic slip represent unambiguous dynamic seismic slip indicators.

3. Comprehensive Option Analysis

Option	Field / Terminology	Scientific Evaluation
(a) Earthquakes of the geological past through slippage observed in the field along faults	**Palaeoseismicity (Paleoseismology)**	**Correct.** Exact definition. Focuses specifically on prehistoric/geological earthquakes deciphered from stratigraphic displacements and fault-trench exposures.
(b) Relationship between earthquake and faulting	**Seismotectonics / Active Tectonics**	*Incorrect.* The general mechanical and geometrical relationship between faults and earthquakes is termed seismotectonics or earthquake fault mechanics, not paleoseismicity.
(c) Mapping of earthquakes	**Seismicity Mapping / Seismic Cartography**	*Incorrect.* Plotting epicenters and focal mechanisms on regional or global maps describes seismicity mapping (e.g., Wadati-Benioff zones, plate boundaries).
(d) Risk related to seismic activities	**Seismic Hazard & Risk Assessment**	*Incorrect.* Quantifying social, economic, or infrastructural damage from potential ground shaking is seismic risk/hazard analysis (Probabilistic Seismic Hazard Analysis, PSHA).

4. Authoritative Textbook References

- **Van der Pluijm & Marshak (2004)** *Earth Structure: An Introduction to Structural Geology and Tectonics*, 2nd Ed., W. W. Norton, Chapter 8 ("Faults and Faulting"), Section 8.7.2 ("Faulting and Earthquakes"), pp. 199–201, Fig. 8.37.
- **Haakon Fossen (2016)** *Structural Geology*, 2nd Ed., Cambridge University Press, Chapter 10 & Chapter 17 (Earthquakes, slip events, and fault rocks).
- **James P. McCalpin (2009)** *Paleoseismology*, 2nd Ed., Academic Press / Elsevier, Chapter 1 ("Introduction to Paleoseismology").
- **P. K. Mukherjee (2013)** *A Text Book of Geology*, World Press, Chapter on Earthquakes and Crustal Movements.